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Current index based protection technique for low voltage DC microgrid

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ABSTRACT

In recent days, the penetration of power electronic converters in microgrid is expecting fast operation of the protection schemes for healthy operation of the power electronic devices. The threshold selection based protection methods are system dependent which may not be reliable. In addition, protection methods based on the principle of overcurrent and reversal of current direction are not preferable for DC microgrid protection. Moreover, protection coordination is one of the challenge in case of failure communication. This article proposes a simple algorithm that utilizes only local current samples to achieve above mentioned challenges. The central controller enhances the reliability of protection system by providing backup operation in case of communication failure. This method adopts polarity of the current index obtained at each end of the line segment in the DC microgrid to recognize and isolates the faulted line promptly. The dissimilar abnormal cases like different operating modes, load changes, faults at different line locations with various system configurations and different fault resistances have been simulated in the MATLAB/Simulink tool to test efficiency of the proposed method. A comparison of the proposed method with existing methods has been made to examine the accuracy in dependability and reliability.

1. Introduction

1.1. Motivation for the research work

Due to the rapid growth of DC loads towards the residential and industrial applications, the integration of microgrids with conventional power systems has been pre-dominated [1,2]. Nowadays, low-voltage direct current microgrids are enhanced due to their asset over alternative current microgrids. The benefits of DC microgrid over the AC microgrid are high efficiency, no concept of phase and frequency control, no synchronization problems, and absence of skin effect [3–5]. The power conversion levels are less since most of the resources and energy storage components are inhered with DC output. The operation of the DC grid is more superior under normal conditions. Meanwhile, the DC microgrid has challenges against protection due to the presence of power converters and the absence of zero-crossing point [6]. In addition, the tremendous change in current that occurs at the instant of fault inception is due to the lower length of line segment and high DC link capacitance across power converters [7]. The insulated gate bipolar transistor based power converters are inherently self-protected and allow the freewheeling diodes to expose high fault currents, which break down the switches of the converter [8,9]. Hence DC microgrid demands a reliable protection method to isolate the fault section before the voltage across DC link capacitor reaches zero or negative [10].

1.2. Literature survey

Many researchers have introduced different protection methods in the literature to overcome the above issues associated with DC microgrids. In [11], the protection scheme relies on the overcurrent strategy to detect the faults in DC microgrids. But, the performance of this scheme may not be effective during high fault resistances. [12] and [13] have proposed the current difference protection technique for medium voltage direct current (MVDC) islanded microgrids. Despite that, this approach requires reliable high bandwidth communication infrastructure to transfer the sensed current data from one end to another end. Moreover, it may not precisely distinguish internal and external faults. The protection method in [14] is committed to the co-efficient of fault current and adjustment of hierarchy of the relay for accurate operation. Nevertheless, the setting of a relay is system dependent. A protection method developed in [15] relies on the concept of slope of fault current for fault detection in DC microgrids. Regardless, this technique may fails to detect faults in case of power reversal, especially in ring configuration microgrids. [16] has introduced fast and reliable protection methods by

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employing the overcurrent approach. But, the operation of this method is not satisfactory in terms of selectivity and relay coordination during low magnitude current faults.

An overcurrent and real-time discrete wavelet transform (RTDWT) strategies have been adopted to detect low and high fault resistances, respectively. However, the operating time of this method to derive trip signal is high in case of high resistance faults [17]. In [18], the protection method adopts the synchronization of measured currents for the detection of faulty line. Nevertheless, this method is not practical in terms of classification of the faults. The protection method is utilized local measurements to compute the difference in current (Δ I) for fault detection in DC microgrids. However, this method works on the principle of threshold setting and it may not precisely detect the faults during low current magnitude faults [19]. In [20], the protection strategy exploited the polarity of fault current detected at intelligent electronic device (IED). Still, this strategy may not be accurate to identify the appropriate fault direction during low current magnitude faults. The frequency of current and inductance parameter have been utilized for fault detection and fault isolation, respectively, in DC microgrid [21–24] have introduced a disturbance index to detect the faults in ring-configured microgrids. However, this method may leads to loss of dependability of the system during radial configuration and high resistance faults. In addition, the operation of these methods is ineffective in case of communication failures. [25] has investigated iterative protection strategy adopts the difference between measured and theoretical fault currents. Regardless, this approach takes more time to derive trip decision due to dependency on the initial assumption of fault location and fault resistance. [26] has presented a centralized protection scheme to detect and isolate faulted line for low voltage microgrids. Nevertheless, the failure of a controller may reduce the reliability since this method is committed with single IED.

1.3. Contributions and organization of the manuscript

This paper proposes a fast and reliable protection method to detect and isolate faulted line under different configurations and operating modes to accomplish the above mentioned challenges in literature. The contributions of the proposed protection method comprises as follows:

1) The proposed method is fast and straightforward, which requires only locally sensed current samples to determine the current index (CI).
2) Subsequently, the coefficient of current index is employed to categorize fault as an internal or external fault.
3) The current index estimated at both the terminals of the line segment holds the positive coefficient for internal faults. On the other hand, for external faults or load disturbances, the estimated index at anyone terminal of the line possesses a negative index coefficient.
4) This method adopted a dedicated communication protocol for effective communication between local intelligent electronic devices (LIEDs) and a central intelligent electronic device (CIED), which will enrich the security and reliability of the system.
5) During the validation process of the proposed method, the dissimilar fault cases like various fault resistance values, faults at different locations in line with various microgrid configurations, different operating modes, and fluctuations in load have been simulated.
6) A comparative analysis of the proposed approach with available methods in the literature has been performed.

Likewise, the organization of this manuscript is demonstrated as follows. Firstly, the concept of fault analysis in the DC microgrid is explained in Section 2. Section 3 discussed the tested microgrid and flow of the proposed protection algorithm. The study of proposed algorithm under numerous fault cases and comparative study with existing methods have been described in Section 4. Subsequently, the conclusions of the proposed work have been provided in Section 5.

2. Fault analysis in DC microgrid

The DC microgrid offers low line impedance as well as high DC link capacitance, which allows the flow of a high magnitude of fault current. Though power electronics switches like IGBT can self-protect against overcurrent, the reverse diodes will let high fault current [27]. In case of a fault, the response of current follows three stages (i) due to discharge of DC link capacitor, (ii) discharge of line inductance through the freewheeling diode and (iii) dynamics of the converter. However, due to the lower withstand capability of converters, it is necessary to isolate the faulty line before completely discharging of DC capacitor.

Case (i)

Discharge of DC link capacitor: In case of a fault, both DC link capacitor and converter feed the fault current. However, immediately during the inception of fault, the contribution of fault current by the converter is neglected. After immediate inception of fault, which establishes a series RLC circuit and the contribution of fault current is due to discharge of DC-link capacitor as shown in Fig. 1a.

During such a case, the response of current is as follows [28]:

$$I(S) = \frac{\frac{V_{DC}(0)}{L} + I_L(0)S}{S^2 + \left(\frac{r}{L}\right)S + \frac{1}{LC}} \tag{1}$$

where, $I_L(0)$ and $V_{DC}(0)$ are initial inductor current and DC link capacitor voltage, respectively. L is the inductance of the line, r is the series combination of line resistance (R) and fault resistance (R_F). The fault current response of (1) in the time domain can be expressed as

$$I(t) = \frac{V_{DC}(0)}{\omega L} * e^{-at}\sin\omega t + I_L(0)e^{-at}\left[\cos\omega t - \left(\frac{a}{\omega}\right)*\sin\omega t\right] \tag{2}$$

The nature of the current response relies on the magnitudes of a^2 and ω_0^2: the fault current response would be under-damped, critical-damped, and over-damped for $a^2 < \omega_0^2, a^2 = \omega_0^2$ and $a^2 > \omega_0^2$, respectively. Due to high DC link capacitance and low impedance of the cable, the initial voltage across the capacitor pre-dominates [29],

$$I(t) \cong \frac{V_{DC}(0)}{\omega L}e^{-at}\sin\omega t \tag{3}$$

The time taken to reach a peak magnitude of fault current in (3) during over-damped and under-damped conditions are [28],

$$t_p^{over} = \frac{1}{\omega}tan^{-1}\left(\frac{\omega}{a}\right)$$

$$t_p^{und} = \frac{\ln(m_2/m_1)}{(m_1 - m_2)} \tag{4}$$

where, $m_1, m_2 = -\frac{r}{2L} \pm \sqrt{\frac{Cr^2-4L}{4CL^2}}, a = \frac{R+R_F}{2L}$, $\omega = (\omega_0^2 - \alpha^2)^{1/2}$ and $\omega_0 = \frac{1}{\sqrt{LC}}$.

Case (ii)

Due to discharge of line inductance: This phase starts when the voltage across the capacitor becomes zero or negative. The line inductance supports the flow of fault current through the freewheeling diode of the converter, as illustrated in Fig. 1b.

The current response can be expressed as,

$$I(t) = \frac{V_D}{r_D + r} + K_3e^{m_1t} + K_4e^{m_2t} \tag{5}$$

From the equivalent circuit in Fig. 1b, the voltage across diode-capacitor will be

$$V_C(t) = \frac{V_D}{r_D + r}r + K_3e^{m_1t}(r + m_1L) + K_4e^{m_2t}(r + m_2L) \tag{6}$$

where, r is the sum of line and fault resistances, $a = \frac{r}{2L} + \frac{1}{2r_DC}, \omega_0 = \frac{1}{\sqrt{LC}} + \frac{r}{\sqrt{(r_DLC)}}$ and $K_{3,\,4}$ are constants.

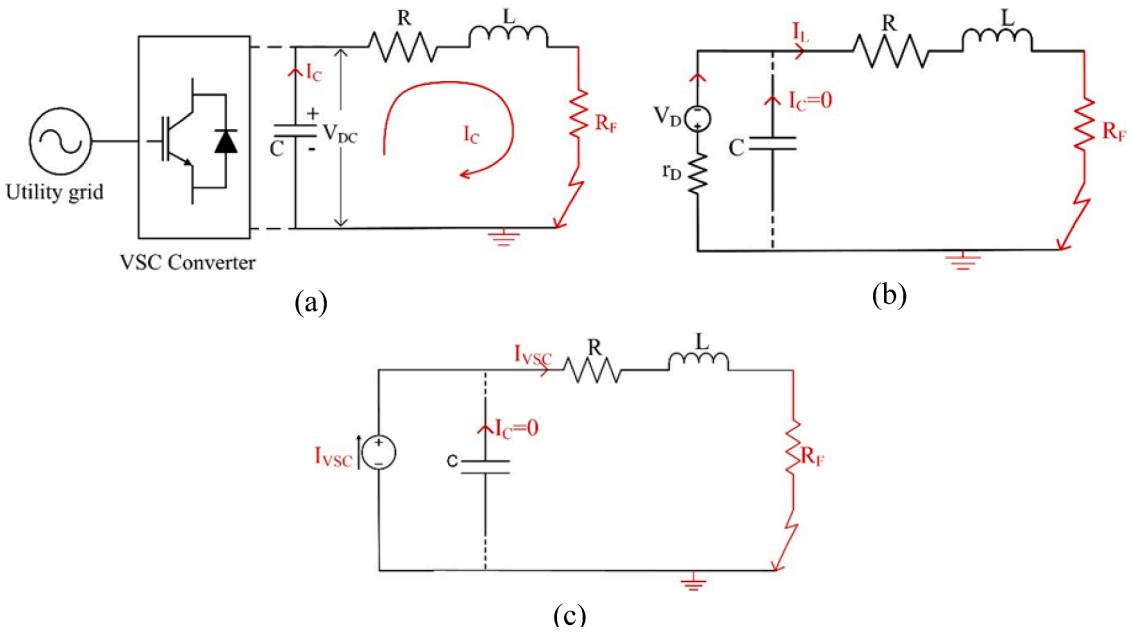


Fig. 1. Equivalent circuit during (a) DC-link capacitor-discharge mode, (b) inductor discharge mode and (c) fault current fed by the converter mode.

Case (iii)

Dynamics of the converter: Immediate after the complete discharge of inductance, the current continues to flow through the converter with the support of a utility grid, as depicted in Fig. 1c. The phase current contributed by the grid during fault condition is

$$i_G^A = I_G \sin(\omega t + a - \beta) + [I_{G0}\sin(a - \beta_0) - I_G \sin(a - \beta)]e^{-\psi\tau} \quad (7)$$

where, $\beta = \tan^{-1}(\omega_0 \tau)$, $\tau = (L_{AC} + L)/r$, β_0 and I_{G0} are initial phase angle and current, respectively. The total fault current contributed by the converter can be expressed as

$$I_{VSC} = i_G^A + i_G^B + i_G^C \quad (8)$$

where, i_G^A, i_G^B and i_G^C are the grid currents of phase A, phase B and phase C, respectively.

3. Proposed protection method

The proposed protection method utilizes only locally measured samples of current instead of remote end current data to derive tripping commands for the appropriate isolation of faulted line. The 350 V, 500 kW tested DC microgrid schematic includes the solar photo-voltaic, wind generator, battery bank, utility grid, and loads. The sources and loads are interfaced through appropriate power converters, as shown in Fig. 2. The wind and solar have employed the MPPT control method to supply power to the microgrid. The tested microgrid is also combined with local intelligent electronic devices (LIEDs) and a central intelligent electronic device (CIED) to improve the selectivity and reliability of the system. At each end of the line holds the current sensor, circuit breaker (CB) and LIED. Each LIED attains the locally measured current samples with the help of current sensor and executes the proposed protection algorithm. Subsequently, generated logic signals only transfer to the

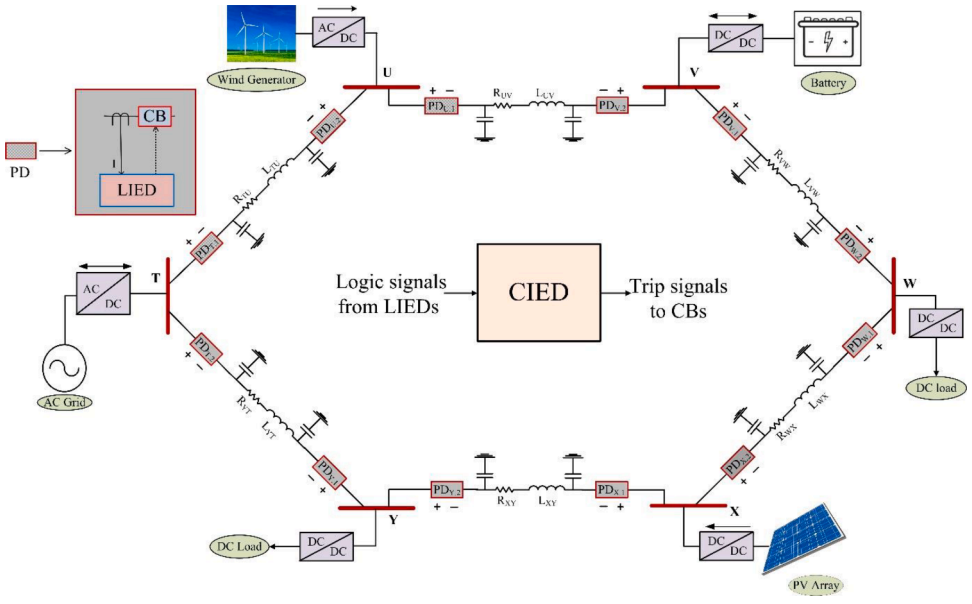


Fig. 2. The architecture of six terminal DC microgrid.

other end local IED. Simultaneously, all LIEDs transfer the generated logic signals to CIED for backup protection. The execution of the protection algorithm at LIED itself will reduce the dependency on CIED. According to the proposed method CIED issue trip operation if there is a loss of communication between LIEDs, communication failure between local and central IEDs, and failure of circuit breakers. The flow of the proposed approach is illustrated in Fig. 3.The effectiveness of the proposed protection algorithm is tested on ring configuration DC microgrid under distinct abnormal conditions in MATLAB/Simulink simulation software. All parameters of the microgrid under the test are given in Table 1.

Due to the low value of DC line impedance and high DC link capacitance, the microgrid exposes to a huge change in current in case of faults. The high magnitude current may spoil power converter switches when the fault current flows through them. This scenario demands a fast operating protection method to provide trip operation to the breakers before DC link voltage becomes negative. The proposed method estimates the current index at each IED at a sampling frequency of 8 kHz. A current index is estimated by employing the following expression:

$$CI = \frac{1}{P}\sum_{q=1}^{P}\frac{I_{q+1} - I_q}{t_{q+1} - t_q} \tag{9}$$

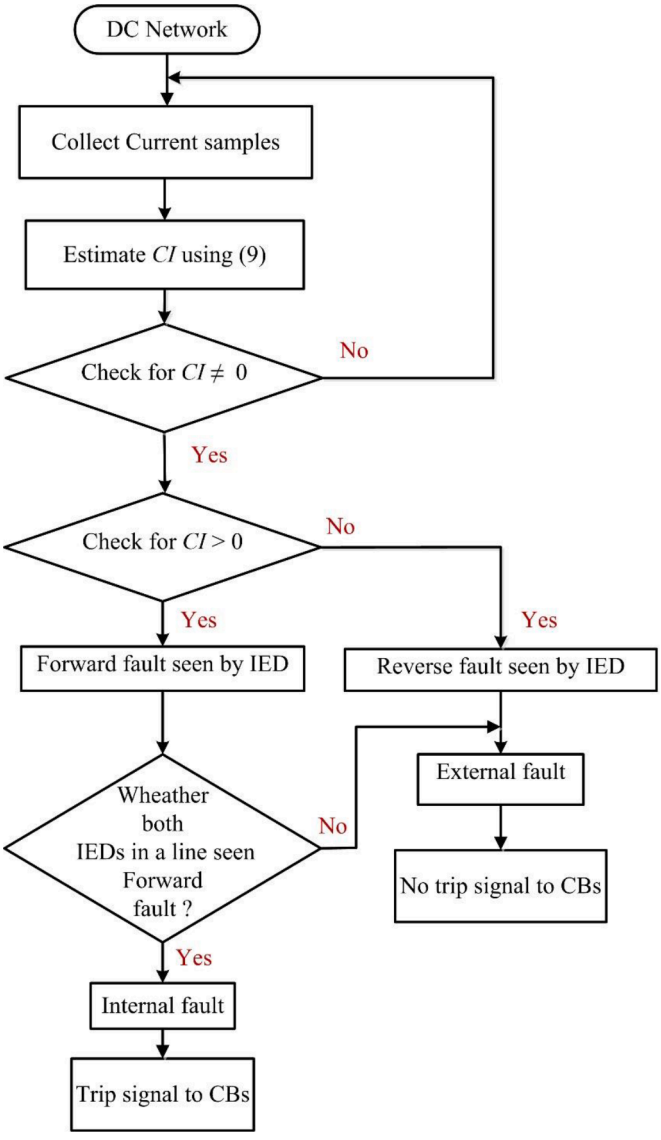


Fig. 3. Flowchart of proposed protection approach.

Table 1
DC microgrid parameters [22,23].

Parameters	Values
System voltage	350 V
Solar panel	V_{mp} = 29 V, I_{mp} = 7.35 A, P_{max} = 213 W
Battery	96 V, 0.4 kAh
Wind turbine	500 kW
Solar converter	125 kW
Battery converter	125 kW
VSC connecting to a utility grid	250 kW
DC link Capacitance	25 mF
Line resistance	121 mΩ/km
Line inductance	970 μH/km
Line capacitance	0.67 μF/km
Load power	500 kW

where, I_{q+1} and I_q are the current samples corresponding to two successive samples of $(q+1)^{th}$ and q^{th}, respectively, $(t_{q+1} - t_q)$ is represents the sample period (T_S) and $p = 4$.

The non-zero value of index will ensure that the microgrid is subjected to the disturbance i.e., fault or load change. Further, the polarity of current index confirms the direction of fault with respect to each LIED. The positive and negative value of index estimated implies that the respective IED has seen the fault in forward and reverse direction, respectively. Subsequently, based on the information received from the other end proposed algorithm discriminates internal and external faults precisely. For internal fault, the current index estimated at both the end IEDs in a line possesses positive values. However, for external fault, either one IED holds a negative current index.

The local intelligent electronic device (LIED) acquire the current samples and executes the proposed protection algorithm in itself. Further, it generates logic signal of '1′ or '0′ and transfer to other end LIED and CIED simultaneously. At the same time, it communicates with other end LIED to issue appropriate trip operation to the respective circuit breakers. The work flow of LIED is depicted in Fig. 4.

As per the proposed algorithm, the CIED (Central intelligent electronic device) enhances the reliability of the system in case of communication failures and failure of circuit breakers. Suppose, if there is a loss of communication between local and central IEDs, the CIED issues a trip signal to the respective breakers based on logic signals received from LIEDs associated with adjacent lines. In case of circuit breaker failure, the CIED refers the status of remote LIEDs to issue trip signal to the remote breaker. Initially, the CIED receives logical signals from all LIEDs and check for pair of LIEDs that holds logic '1′. If pair of LIEDs hold logic '1′ then the CIED issues trip operation to the concerned breakers through the respective LIEDs. The work flow of CIED is given in Fig. 5 and also presented below for clarity. The communication delay between LIEDs and CIED as (1–2)ms.

3.1. Communication between LIEDs

The communication between local intelligent electronic devices is essential to examine protection algorithm in terms of reliability and speed of operation. The communication arrangement between the LIEDs positioned in a line is illustrated in Fig. 6.

According to the proposed algorithm, LIED obtains the locally measured current samples in place of remote data. Afterwards, the proposed algorithm executes at both LIEDs in a line and the logic signal is generated. If the estimated current index is positive, it means that the fault occurred in the forward direction to the respective LIED and then it generates logic signal '1′. However, the negative value of current index indicates that fault is in the reverse direction and LIED issues logic '0′. After that, the logic signal is generated at one end and communicates with the other end to identify and isolate faulted line and enhance the selectivity by distinguishing internal and external faults. However, there may be a chance of communication failure between local IEDs for

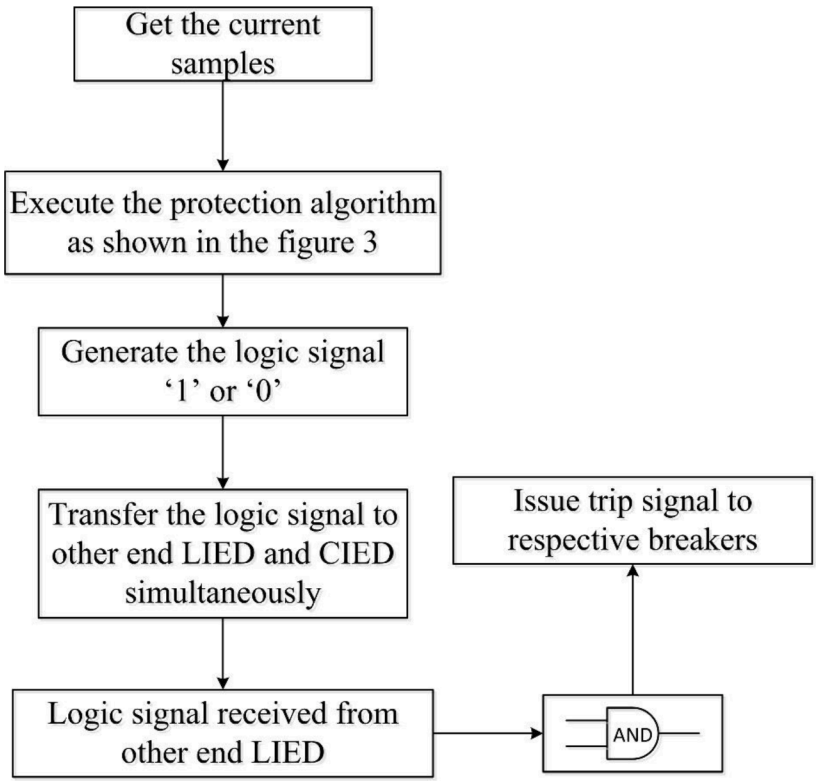


Fig. 4. Work flow of local intelligent electronic device (LIED).

various reasons. Under these circumstances, the CIED provides backup trip operation to enhance the reliability and dependability of the protection method. The different case studies of communication failure have been discussed as follows:

3.2. Failure of communication between local IEDs

The reliability of the system may degrade due to the failure of communication between local IEDs. Under this condition, the central IED issue appropriate trip operation based on logic signals obtained from respective LIEDs pertaining to the respective line segment. During a fault in the UV line, the $LIED_{U.1}$ and $LIED_{V.2}$ will generate a logic signal. Based on the communication, both LIEDs issue the trip operation to the respective circuit breakers, as shown in Fig. 7. Nevertheless, in case of communication failure between $LIED_{U.1}$ and $LIED_{V.2}$, then the CIED simultaneously issues trip operation to the respective breakers.

3.3. Failure of communication between local IED and central IED

Suppose, there is a loss of communication between local and central IEDs. In that case, the CIED issues a trip signal to the respective breakers based on logic signals received from LIEDs associated with adjacent lines, as depicted in Fig. 8. For instance, if there is a communication failure between $LIED_{U.1}$ and CIED. The CIED will make a decision based on the data received from $LIED_{V.2}$ and $LIED_{T.1}$ and issue the trip signal to $CB_{U.1}$ and $CB_{V.2}$ simultaneously. It ensures the reliability of the protection scheme.

3.4. Failure of breaker

The failure of a circuit breaker in a particular faulted line results in no operation of local communication between IEDs associated with the faulty line and also between respective LIED and CIED. During such case, the CIED communicates with the LIED associated with an adjacent faulty line and simultaneously issues trip operation to the respective breakers, as depicted in Fig. 9. In the case of $CB_{U.1}$ failure, the CIED communicates with $LIED_{T.1}$ and $LIED_{V.2}$ and provides trip operation to both $CB_{T.1}$ and $CB_{V.2}$ for the isolation of TU and UV lines.

Therefore, it is recognized that the proposed protection scheme will strengthen the reliability and dependability of the microgrid during the communication failures and/or failure of CB by ensuring the backup operation of breakers.

4. Results and discussion

4.1. Enhancement of dependability

The dependability of the protection system would be ensured by validating the proposed protection algorithm under various circumstances such as multiple fault locations with different configurations, different values of fault resistance, different operating modes and the absence of wind farm.

4.1.1. Various fault locations under different configurations

During the validation process, a line–ground (LG) fault with a fault resistance value of 1 Ω at multiple locations in the TY line at $t = 0.2$ s under different configurations is considered. In this contrast, the line UV is detached from the microgrid. Under this fault condition, the measured currents during both the configurations at $IED_{Y.1}$ and $IED_{T.2}$ have shown in the Fig. 10.

From this fault study, it is evinced that the fault current value decreases when the fault move towards the other end. Additionally, the magnitude of current in the TY line during radial configuration is slightly higher than the current value as in the case of ring configuration. For all fault locations, the current index at $IED_{Y.1}$ and $IED_{T.2}$ have been estimated and listed in Table 2.

The estimated current index values at $IED_{Y.1}$ and $IED_{T.2}$ have positive nature for various fault locations in the TY line under different configurations. Consequently, both IEDs see the fault in the forward direction and declare an internal fault. Hence, the proposed algorithm correctly

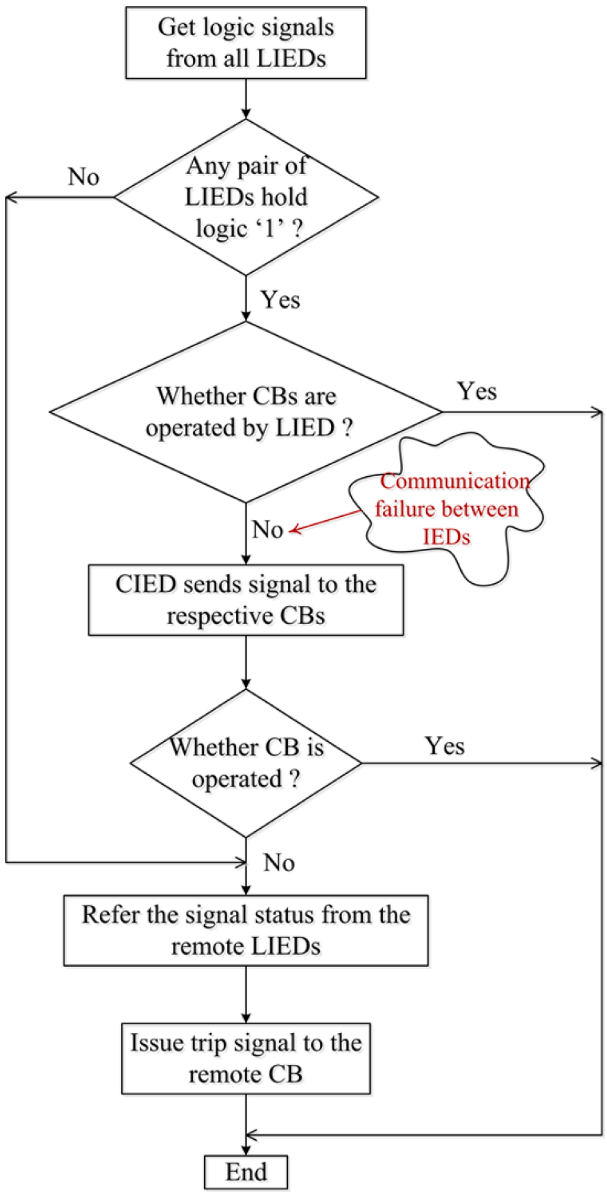


Fig. 5. Work flow of central intelligent electronic device (CIED).

identify and isolate the faulted line even during change in the configuration.

4.1.2. Different fault resistance values

To examine effectiveness of the proposed technique, a fault is simulated at $t = 0.2$ s in the VW line with different fault resistance values at 200 *m* away from bus V. During such case, measured current at $IED_{V.1}$ and $IED_{W.2}$ as exhibited in the Fig. 11.

It is found that the level of fault current varies with the magnitude of fault resistance. During such fault cases, the current index values determined at $IED_{V.1}$ and $IED_{W.2}$ have listed in Table 3.

Table 3 implies that as fault resistance increases, the value of estimated index decreases. For all fault resistances, the estimated index at $IED_{V.1}$ and $IED_{W.2}$ possess positive values. As a result, the proposed method accurately isolates the VW line and dependability has been ensured.

4.1.3. Different operating modes

Due to the natural characteristics of resources, the DC microgrid needs to operate in islanding and grid-connected modes for effective power management. Consequently, the microgrid is subject to different fault current levels during fault conditions. To test the proposed protection algorithm under this challenging condition, a LG fault in UV line 300 m distance from bus V with fault resistance of 0.5Ω has been simulated.

The corresponding currents measured at $IED_{U.1}$ and $IED_{V.2}$ are shown in the Fig. 12. During this fault study, the estimation of current index at $IED_{U.1}$ and $IED_{V.2}$ have given in Table 4.

Table 4 shows that the estimated current index at $IED_{U.1}$ and $IED_{V.2}$ holds positive value under both the modes of operation. Hence, it is observed that the performance of the proposed algorithm is not influenced by the grid operating modes.

4.1.4. Absence of resource

The performance of protection algorithm is also evaluated to test the dependability in the absence of wind generator. During the evaluation process, the inception of a LG fault 700 m away from bus V with R_F of 5 Ω is considered. The currents measured at respective $IED_{V.1}$ and $IED_{W.2}$ are shown in the Fig. 13.

It implies that the microgrid experience different fault currents due to the absence of wind generation. For this case study, the estimated current index at $IED_{V.1}$ and $IED_{W.2}$ have provided in Table 5. It is revealed that the index value estimated at $IED_{V.1}$ and $IED_{W.2}$ in both the cases maintains a positive index. Subsequently, both IEDs identify the fault in the forward direction and corroborate that the fault is internal. Hence, it is evident that the intermittent nature of the wind generator does not influence the current index based algorithm.

4.2. Enhancement of selectivity

To examine the impact of fault in other line segments, LG fault in the TU line at a distance of 500 m from bus T with R_F of 0.01 Ω is simulated. During this fault condition, the current index estimated at all IEDs using (9) has listed in Table 6.

The results show that both IEDs positioned in faulted line have positive values of the current index. At the same time, either one of the IED in healthy lines would hold negative value. As per the proposed protection algorithm, the positive value of the index indicates that the fault is located in the forward direction concerning IED; otherwise, the fault is a reverse fault. Moreover, the IEDs pertaining to faulted line TU carries positive values and apprised that the fault is internal; otherwise, it is an external fault. Hence, the proposed technique perfectly distinguishes the internal and external faults and enhances the selectivity of the protection devices.

4.3. Enhancement of security

The efficiency of proposed algorithm is also investigated in terms of enhancement of security of the protection device. For this condition, the various load switching cases such 80%, 100%, 120% and 130% have been simulated. The corresponding measured currents are shown in the Fig. 14.

Under load switching conditions, values of the estimated index at $IED_{V.1}$, $IED_{W.2}$, $IED_{W.1}$ and $IED_{X.2}$ have listed in Table 7. It is understood that either one of IED pertaining to the line segments VW and WX would maintain the negative value of current index. Consequently, the proposed algorithm treated the sudden switching in load as an external fault. However, all the IEDs have zero current indexes during normal loading conditions. Therefore, the proposed method exactly discriminates the inception of fault and change in load, as well as security of the system has been ensured.

4.4. Comparative study

To examine the effectiveness of proposed scheme, the comparative study with the existing protection schemes in literature has been done.

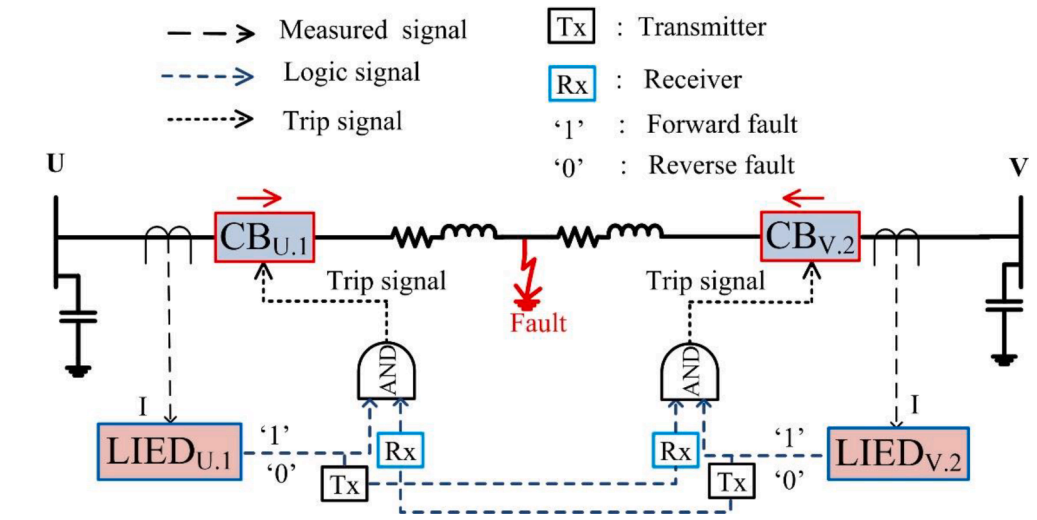


Fig. 6. The communication arrangement between the LIEDs.

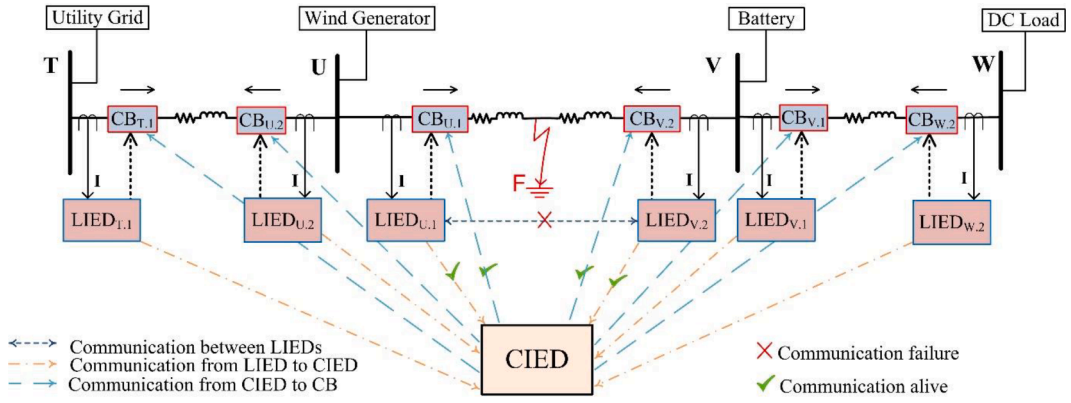


Fig. 7. The failure of communication between local IEDs.

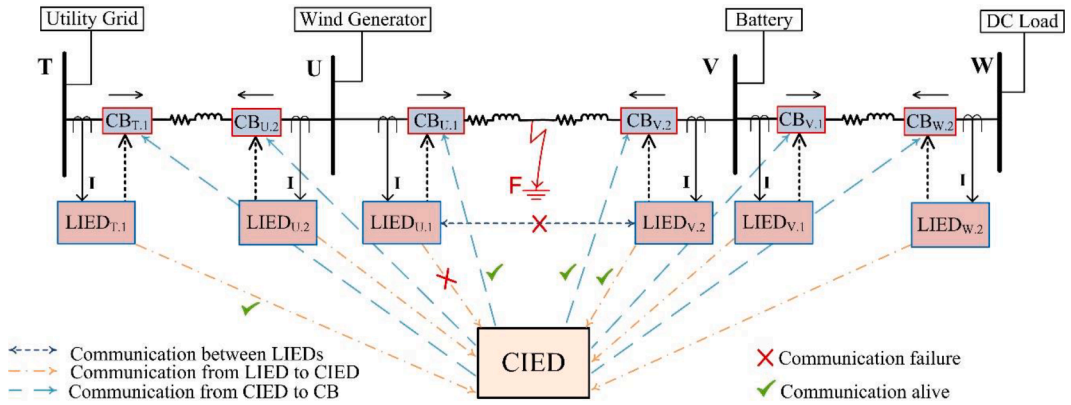


Fig. 8. The failure of communication between local IED and central IED.

4.4.1. Polarity of the current during high resistance faults [20]

The protection method for DC microgrid presented in [20] relies on the polarity of current detected at both the end IEDs. Subsequently, the trip decision would be derived to respective circuit breakers. According to the method in [20], the line will isolate only if the polarity of measured current at respective IEDs is positive. Nevertheless, this method may not correctly distinguish the forward and reverse faults during high fault resistances. To evaluate the superior performance of the proposed method over the existing method a fault study has been done. L-G fault is simulated in the TU line at a distance of 0.4 km from bus T by varying fault resistance values. The comparative assessment of both the methods is given in Table 8.

From Table 8, as IED$_{U.2}$ is the observation point, the sign of current is negative when the fault resistance exceeds 1 Ω and fails to recognize the correct fault direction. Despite that, according to the proposed method the estimated index at IED$_{T.1}$ and IED$_{U.2}$ have positive values even during high resistance faults. Hence, the proposed current index based protection method perfectly detects and isolates the faulted line and

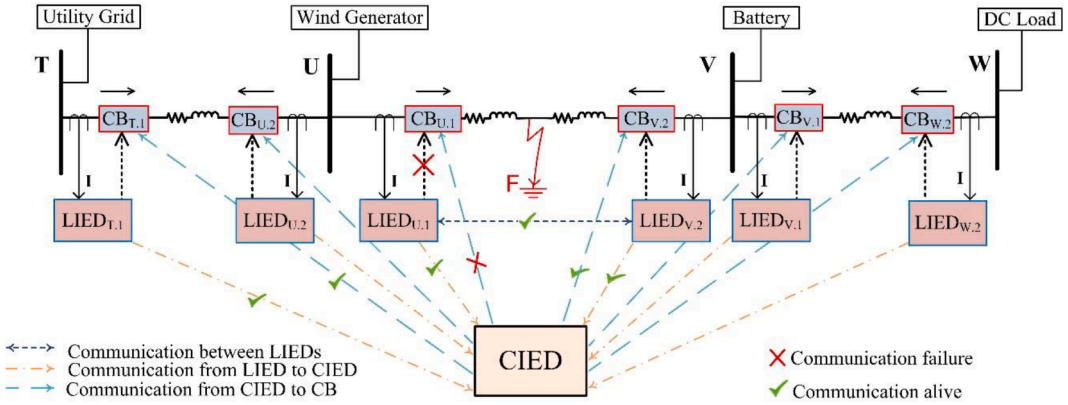


Fig. 9. The backup operation in case of breaker failure.

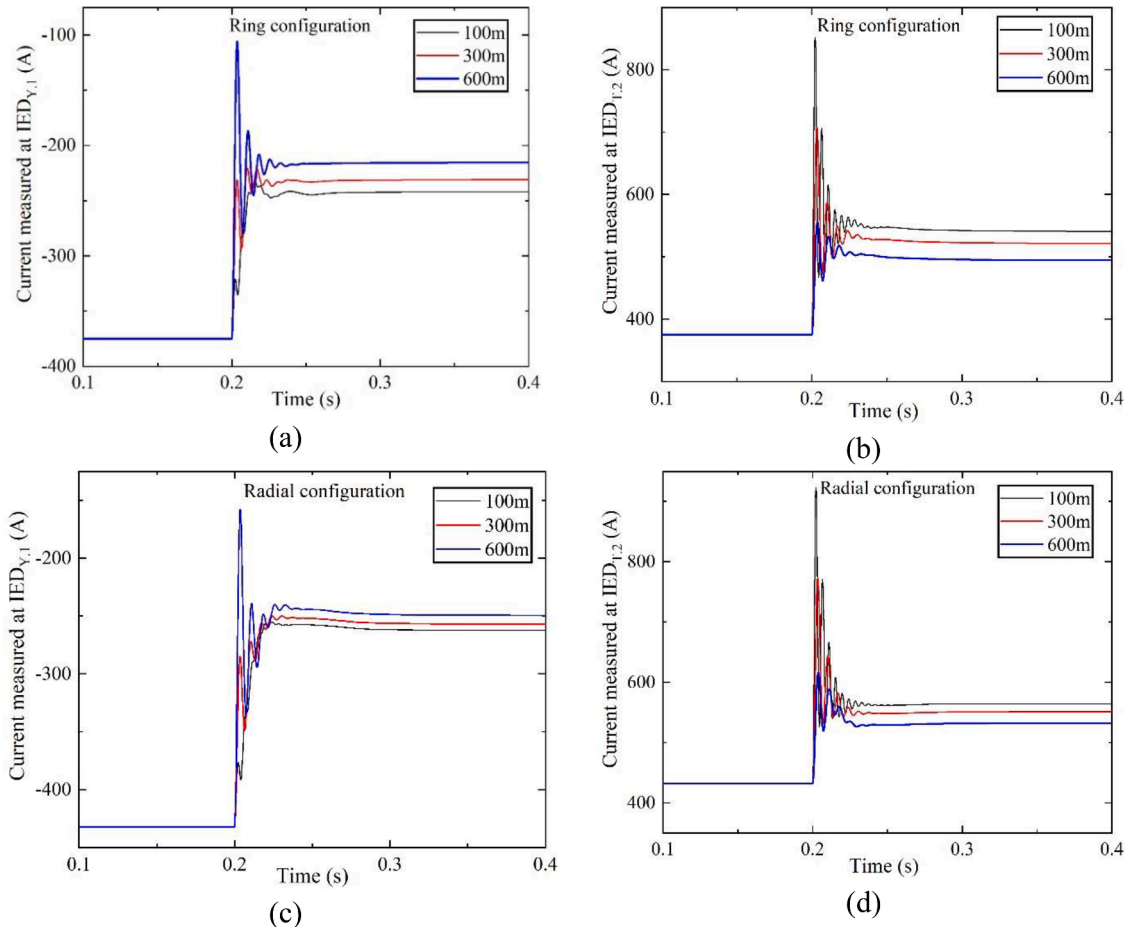


Fig. 10. Current measured during ring configuration at (a) $IED_{Y.1}$ (b) $IED_{T.2}$ and during radial configuration at (c) $IED_{Y.1}$ (d) $IED_{T.2}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

dependability has been ensured.

4.4.2. *Threshold of ΔI [19]*

The efficiency of proposed current index based method is also analyzed with the method in [19]. According to [19], the IED detects the fault only if the estimated ΔI is above the threshold value of 5 A.

For the comparative study, a fault in the WX line for different fault resistances has been considered. The results of two methods have tabulated in Table 9. According to [19], estimated ΔI at $IED_{X.2}$ is not well above the threshold value when the fault resistance is greater than 1 Ω and fails to detect the faults. Regardless, the proposed method adopts the

Table 2
Estimated current index values for different locations and configurations.

Type of configuration	Fault location from bus T	Estimated current index (kA/s)	
		$IED_{Y.1}$	$IED_{T.2}$
Ring	100	13.33	132.96
	300	17.55	43.19
	600	30.73	19.9
Radial	100	13.71	136.8
	300	18	44.29
	600	31.35	20.35

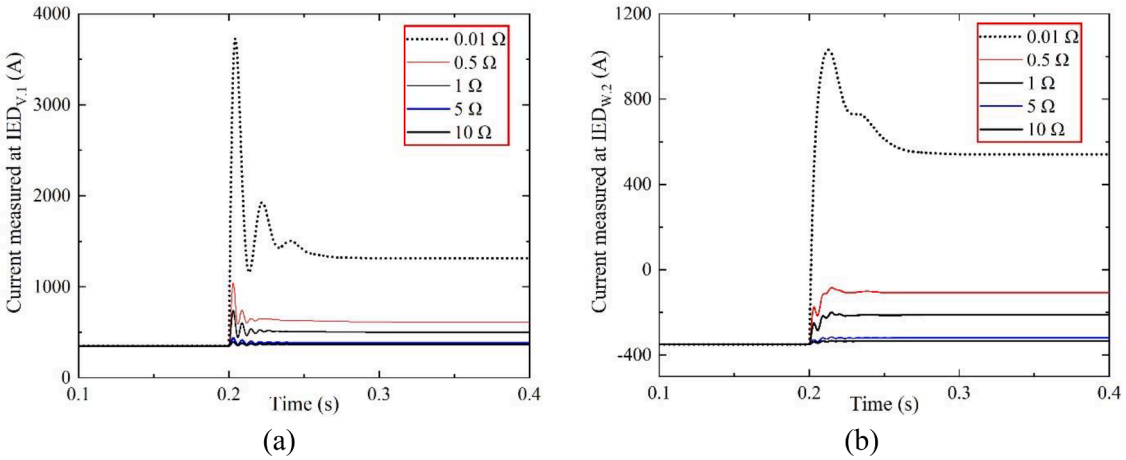


Fig. 11. Current measured under different fault resistance values at (a) IED$_{V.1}$ (b) IED$_{W.2}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Estimated current index values for different fault resistance values.

Fault resistance (R_F) in Ω	Estimated current index (kA/s)	
	IED$_{V.1}$	IED$_{W.2}$
0.01	1246.25	288.4
0.5	125.57	29.03
1	65.09	15.05
5	13.4	3.1
10	6.73	1.55

polarity of current index is estimated using (9). The current index at IED$_{W.1}$ and IED$_{X.2}$ have positive values and meticulously classifies internal and external faults. It ensures the dependability enhancement and reliability of the system.

4.4.3. *Polarity of the current during pre-fault load cases [20]*

A comparative study has been made to examine the genuineness of the proposed method with the protection method introduced in [20] under numerous pre–fault load cases. During this study, a fault in the VW line at various locations with a fault resistance of 0.5 Ω has been simulated. The observations of both the methods at IED$_{V.1}$ and IED$_{W.2}$ are listed in Table 10.

According to the method presented in [20] for 50% load condition,

Table 4
Estimated current index at IED$_{U.1}$ and IED$_{V.2}$ during grid and islanding modes.

Operating mode	Estimated current index (kA/s)	
	IED$_{U.1}$	IED$_{V.2}$
Grid	87.47	35.32
Islanding	76.69	30.96

the current measured at IED$_{W.2}$ holds a positive sign and precisely identifies the fault direction to isolate faulty line. However, when the loading is greater than the full load under various fault locations, the current at IED$_{W.2}$ has a negative sign and sees the fault in reverse direction even though the fault is internal. As per the proposed method, the estimated current index at IED$_{V.1}$ and IED$_{W.2}$ maintain positive values and see the fault is forward for all the fault cases, as given in Table 10. Hence, the proposed method correctly isolates faulty line and improves the dependability of the protection device.

4.4.4. *Time of operation of protection methods*

The DC microgrid expects fast operation of protection techniques to prevent the power electronic devices from high currents during faults. A fault study has been performed to compare the speed of operation of the proposed protection technique with the protection methods introduced in the literature. From this fault analysis, it was found that the operating

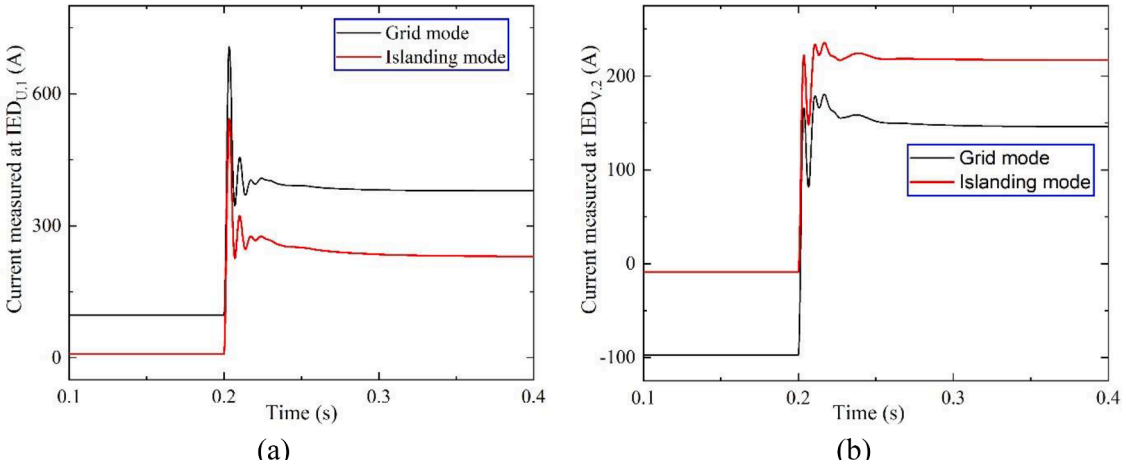


Fig. 12. Current measured under different operating modes at (a) IED$_{U.1}$ (b) IED$_{V.2}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

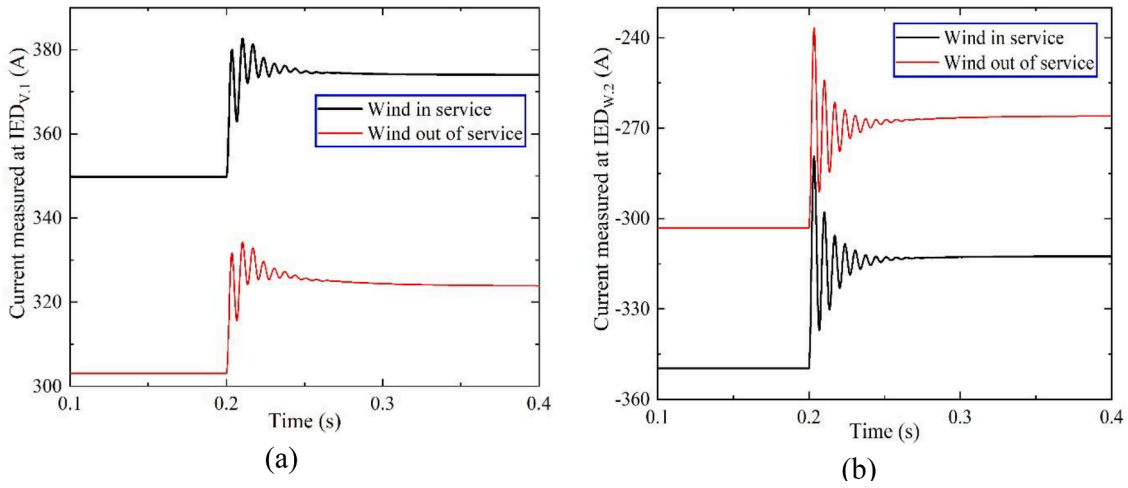


Fig. 13. Current measured with and without wind turbine at (a) IED$_{V.1}$ (b) IED$_{W.2}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Estimated current index with and without operation of the wind turbine.

Operation of wind turbine	Estimated current index (kA/s)	
	IED$_{V.1}$	IED$_{W.2}$
In service	3.37	8.31
Out of service	3.18	7.85

Table 6
Estimated values of the current index at all IEDs and trip command signals.

Line (PQ)	Estimated current index (kA/s)		Fault type	Trip command signals to corresponding CBs
	IED$_{P.1}$	IED$_{Q.2}$		
TU	**986.34**	**982.87**	**Internal**	✓
UV	-620.62	204.164	External	×
VW	-52.7	24.9	External	×
WX	-11.8	-15.7	External	×
XY	31.4	-56.8	External	×
YT	197.3	-612.4	External	×

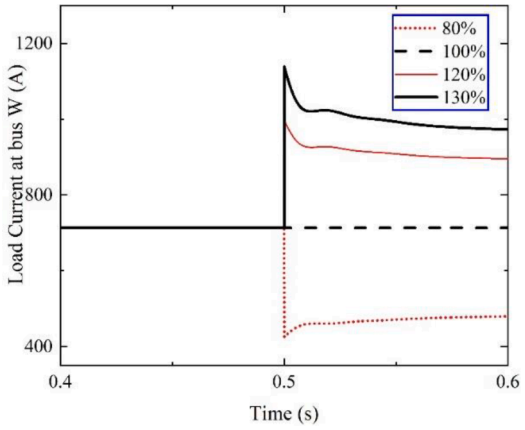


Fig. 14. The current measured at bus W. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

time to issue a trip signal for the operation of the circuit breaker was 0.5 ms in the proposed method. The operating time of different protection methods available in the literature has been provided in Table 11.

Table 7
Estimated current index values under various load conditions.

Loading (%)	Estimated current index (kA/s)			
	IED$_{V.1}$	IED$_{W.2}$	IED$_{W.1}$	IED$_{X.2}$
80	-7.7	8820	24,350	-443
100	0	0	0	0
120	8.2	-9320	-25,720	470
130	12.3	-13,960	-38,522	704

Table 8
Comparison of the proposed method with an existing method in [20] during high resistance faults.

Fault resistance value in Ω	Measured current (A) [20]		Estimated current index (kA/s) in proposed method	
	IED$_{T.1}$	IED$_{U.2}$	IED$_{T.1}$	IED$_{U.2}$
0.05	901.4 (FF)	656.8 (FF)	384.1 (FF)	249.3 (FF)
0.1	759.5 (FF)	520.7 (FF)	249.5 (FF)	161.9 (FF)
1	259.1 (FF)	44.6 (FF)	33.4 (FF)	21.7 (FF)
2	186.8 (FF)	-23.4 (RF)	17 (FF)	11 (FF)
5	138.9 (FF)	-69.9 (RF)	6.8 (FF)	4.4 (FF)
10	121.5 (FF)	-86.5 (RF)	3.4 (FF)	2.2 (FF)
15	115.6 (FF)	-92 (RF)	2.3 (FF)	1.4 (FF)

FF – Forward Fault RF – Reverse Fault.

Table 9
Comparison of the proposed method with an existing method in [19].

Fault resistance in Ω	Estimated ΔI (A) [19]		Estimated current index (kA/s) in the proposed method	
	IED$_{W.1}$	IED$_{X.2}$	IED$_{W.1}$	IED$_{X.2}$
0.01	5434.2	114.7	2800	25.8
0.6	241.5	12.3	122	0.9
1	146.2	5.4	77.34	0.58
2	73.4	2.4	38.9	0.34
5	29.6	1.9	15.6	0.13
10	14.8	1.3	7.85	0.069
15	9.8	0.7	5.23	0.046

From this comparative studies, it is comprehended that the operating time of proposed method is comparatively low compared to existing methods.

Table 10
Comparison of the proposed method with an existing method in [20] during pre-fault load conditions.

Load (%) at bus W	Fault location from bus V (km)	Measured current (A) [20]		Estimated current index (kA/s) in proposed method	
		$IED_{V.1}$	$IED_{W.2}$	$IED_{V.1}$	$IED_{W.2}$
50	0.5	501.2 (FF)	41.15 (FF)	53.53 (FF)	25.39 (FF)
110	0.3	625 (FF)	-118.6 (RF)	80.31 (FF)	1.21 (FF)
110	0.5	593.5 (FF)	-131.3 (RF)	45.76 (FF)	22.63 (FF)
110	0.6	552.4 (FF)	-171.5 (RF)	37.04 (FF)	38.16 (FF)
120	0.4	623.1 (FF)	-129.4 (RF)	57.48 (FF)	10.79 (FF)
120	0.7	559.8 (FF)	-193.1 (RF)	30 (FF)	59.66 (FF)

FF – Forward Fault RF – Reverse Fault.

Table 11
comparison of the different methods with the proposed method in connection with operating time.

Proposed method	[11]	[17]	[20]	[22]	[10]	[30]
0.5 ms	3 ms	5 ms	1 ms	2 ms	2 ms	10 ms

5. Conclusion

In this paper, a dedicated protection scheme for the ring and radial type DC microgrids has been presented. This method adopts only current samples to derive trip signals and reduces computational analysis. Firstly, the current index is computed at each local IED. Subsequently, the coefficient of an index is employed to generate the logic signal, which transmitted to other end LIED and CIED. In case of the communication link failure or delay in communication, the central IED enables the appropriate protection device as backup protection. The assets of the proposed method are time to derive trip signal is more diminutive, effective discrimination of internal or external faults, enhancement of selectivity, reliability, and dependability of protection mechanism. The simulation studies concluded that the proposed method accurately isolates the exact faulted line under different fault conditions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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